more players into the market. "I look at open source as a great enabler for the Indian ecosystem to develop," says Manish.

Transforming industries

Dr Inder S. Gopal, CEO, Indian Urban Data Exchange, says there is evidence that open source can transform an industry. "If we look at data centre networking at the time of SDN and NFV, it was dominated by proprietary solutions, whereas now, when the boxes that have been deployed there are looked at, 80% of them are white box solutions, running for the most part open source software. The open source commercialisation has been driven by the availability of open white box hardware alternatives and mature open source offerings, and that has really transformed the data centre," he explains.

Gopal expects to see a similar kind of transformation in the telco space as well, though it might take a little longer. "I think telcos always move a little bit more slowly than other sectors, but it definitely will happen and there is evidence that it can happen."

Driving factor: Cost or competition?

O-RAN based deployments are becoming increasingly popular. For an operator, the cost effectiveness may be a primary thing, but there is also competition because of the low ARPU. Hence you need to match the infrastructure cost in line with that. The question arises, is it the cost reductions or vendor lock-in that is the driving factor? Another thought-provoking factor is that, if infrastructure costs are brought down, then it is obvious there would be competition, because people could jump in and start playing that role.

Disaggregation is happening and OEM layers are increasing. So, if we split the whole structure from its present monolithic form, system integrators would be needed to play the role, which could create lot more challenges.

But the driving factor in open source is the cost of the solution and the diverse supply chain. It is believed that by using open source we can reduce cost as there is nobody to charge royalty or any fees. But open source is not entirely free, it has its own costs involved, explains Manish.

Open source is no longer about being free, or most viable, or an obvious option, but is pretty much a reality with the move towards the Linux Foundation Networking Fund.

Open source helps get more players and innovation into the play. Once you have the vendor lock-in, you have everything that you need to develop from the vendor. Whatever your requirement, you have a set of engineers who are catering to 15 different customers across the world who have 15 different requirements.

Manish says, "If my solution doesn't get priority in my partner's ecosystem, I am really stuck and will not be able to service my customer. So, fundamentally, when you disaggregate, you unlock and remove this lock-in, opening up for more innovation. You deploy the features faster and in some way be able to monetise."

The cost can be brought down one way or the other through efficiency in deployment, or in optics, or in capital expenditures (CAPEX). Though it is not as simple as it sounds, it is evident from the other sectors that whenever you open up things, it basically brings down the total cost of ownership (TCO), Manish adds.

Subodh says, "We just want to clarify the vendor lock-in proprietary aspect of the OEM level of Cisco, but at the moment, when you look at the intelligent programmable infra and architecture building blocks with open source, ONAP, policy design creation, dashboarding, external APIs, orchestration, network components, VNS, virtual functions infra monitoring, it is now an abundant offering."

While Subodh points out innumerable use cases of ONAP that can design, create, orchestrate, and automate everything in 5G, he also tells telco operators to be aware that code repositories are no longer a lock-in aspect. It's not just about 10,000 people contributing, but a function of how the whole wheel aligns to the common cause of the use case.

"I think I can see that balance. A lot of vendors and telcos are working with Telstra, COX, Orange, Charter, Bell, AT&T for example and have evolved from lab ONAP networks to ONAP reference architectures, and that epochal shift is a very good science," adds Subodh.

Leveraging 5G from core to edge

A variety of workloads have a variety of use cases, and there arises the doubt how end-to-end service orchestration can happen? It could be like wireless networks or CNF/VNF combinations. The end-to-end service stabbing from your customers service is getting translated.

Going into the specifics of slice